HiP-LR: Hierarchical Planning with Localized Failure Recovery for Long-Horizon Multi-Robot Tasks

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Background

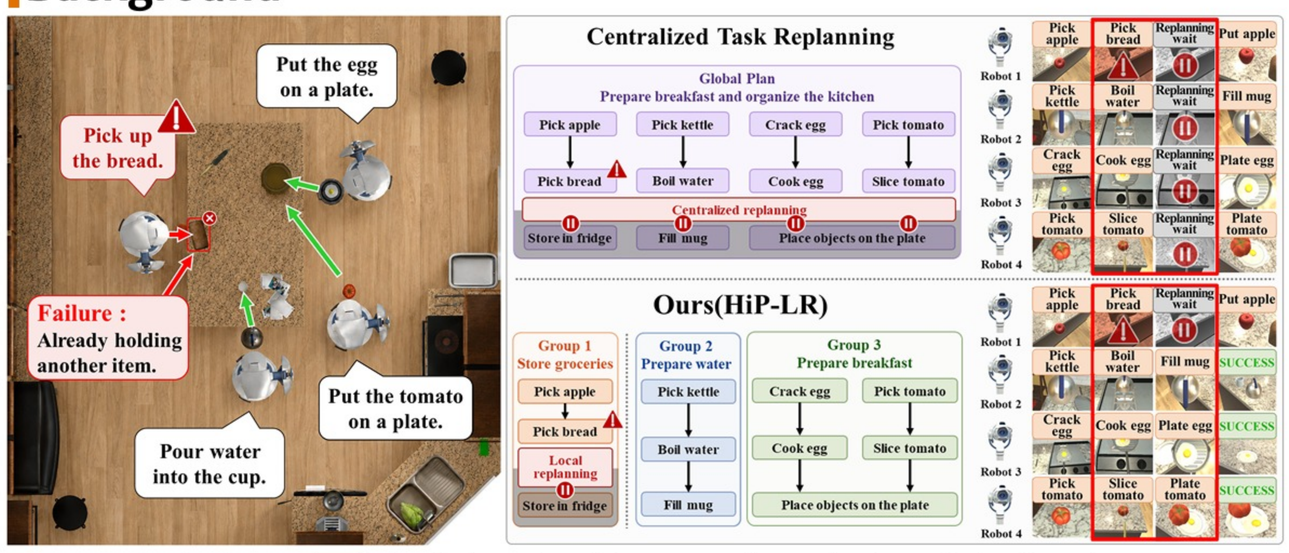
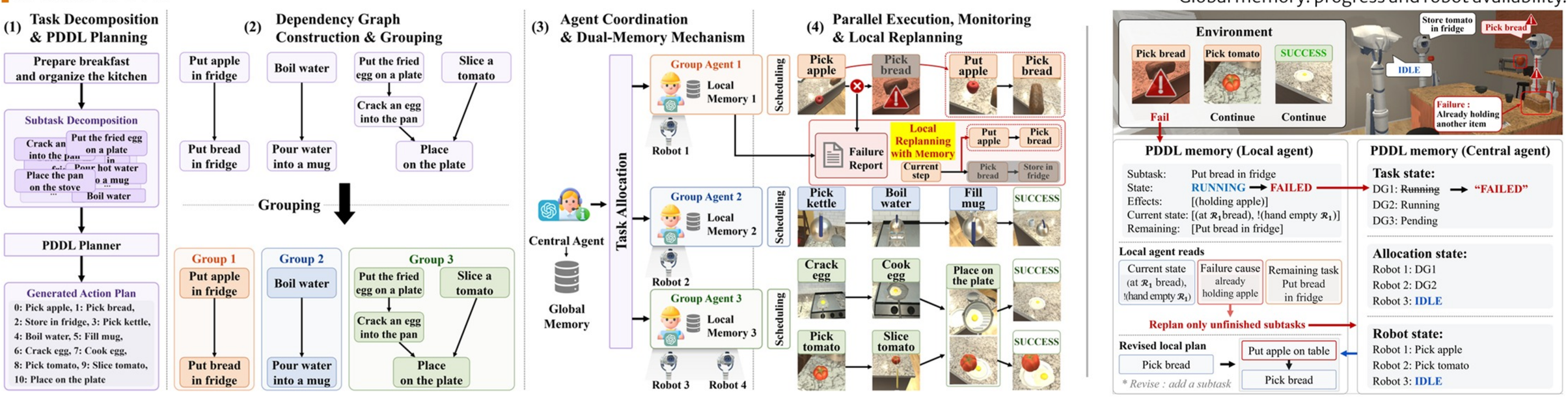


Fig. 1. Centralized replanning (top) halts all robots after a failure. HiP-LR (bottom) replans only the affected group.

LLM-driven multi-robot systems struggle with **long-horizon tasks**. As task dependencies grow, centralized planners suffer from **context explosion**, and a minor local failure forces **every robot to halt during global replanning**, including those unrelated to the failed subtask.

To address this, we propose **HiP-LR**. It divides a task into **dependency-isolated subtask groups**, so a failure pauses only the affected group while the others keep running.

Framework



Local memory: group state and history.
Global memory: progress and robot availability.

HiP-LR groups subtasks by dependency: the Central Agent allocates groups, and Local Agents schedule, monitor, and replan each group.

$$F_k = \langle T_{\text{fail}}, r_{\text{cause}}, H_{\text{action}}, T_{\text{remain}}, s_k^{\text{curr}} \rangle$$

Only the failed group pauses and replans its unfinished subtasks from a PDDL failure report.

Quantitative Results

MAP-THOR					
Method	**SR↑**	**TR↑**	**C↑**	**B↑**	**SE↓**
SMART-LLM	0.04	0.12	0.25	0.09	1.00
DMRS-1D	0.23	0.53	0.76	0.61	0.54
DMRS-2D	0.27	0.53	0.76	0.67	0.52
LaMMA-P	0.30	0.50	0.54	0.34	0.57
LaMMA-P+‡	0.46	0.65	0.69	0.33	0.60
H-AIM‡	0.53	0.66	0.66	0.57	0.75
KGLAMP‡	0.51	0.74	0.76	0.64	0.71
HiP-LR (Ours)‡	0.73	0.90	0.87	0.75	0.48

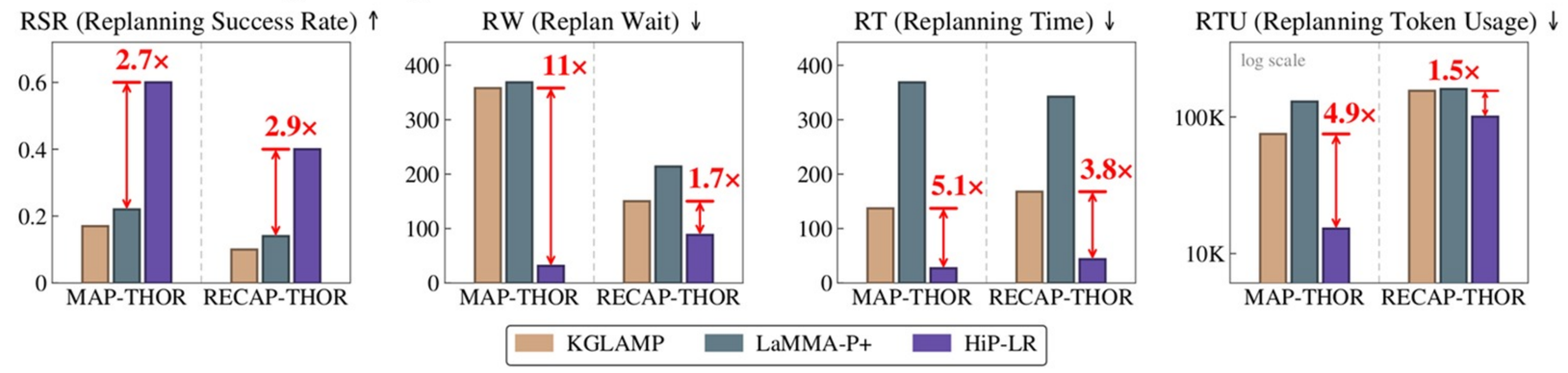
RECAP-THOR					
Method	**SR↑**	**TR↑**	**C↑**	**B↑**	**SE↓**
LaMMA-P+‡	0.25	0.50	0.56	0.21	0.72
KGLAMP‡	0.12	0.60	0.65	0.10	0.66
HiP-LR (Ours)‡	0.47	0.84	0.83	0.58	0.68

‡ Approaches with failure recovery capabilities.

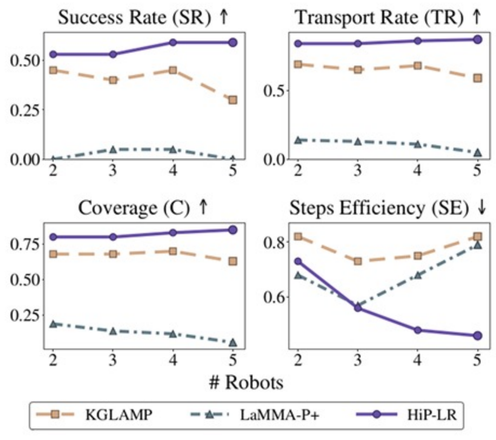
Ablation Study

Method	SR↑	RW↓	RSR↑	RTU↓
w/o Dual Memory	0.62	186.43	0.38	36.3k
w/o Dependency Graph	0.64	74.69	0.14	144.6k
w/o Hierarchical	0.65	57.20	0.50	37.1k
HiP-LR	0.73	34.71	0.90	11.2k

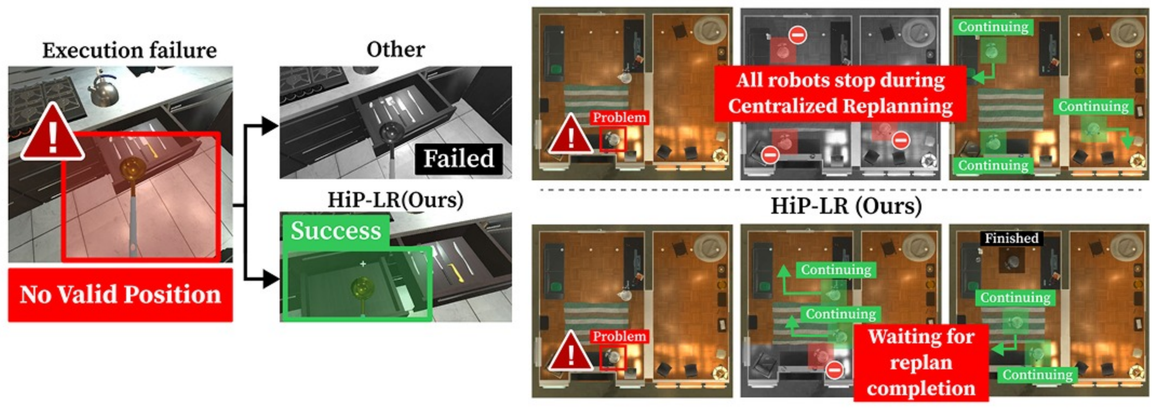
Failure Recovery Analysis



Scalability comparison



Qualitative Results



Group isolation analysis

Config	Tasks	Subtasks	Groups	Mean	90% CI (%)
1	18	51	2.11	−0.010	[−1.7, +0.3]
2	14	65	3.86	+0.001	[−1.1, +0.7]
3	10	37	2.40	−0.021	[−3.3, −1.5]
4	7	28	4.00	−0.015	[−2.9, −1.2]
All	49	181	3.07	−0.010	[−1.6, −0.6]

Conclusion

Partitioning the subtask graph into PDDL-derived dependency groups lets a Central Agent handle global allocation while Local Agents manage scheduling and recovery.

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